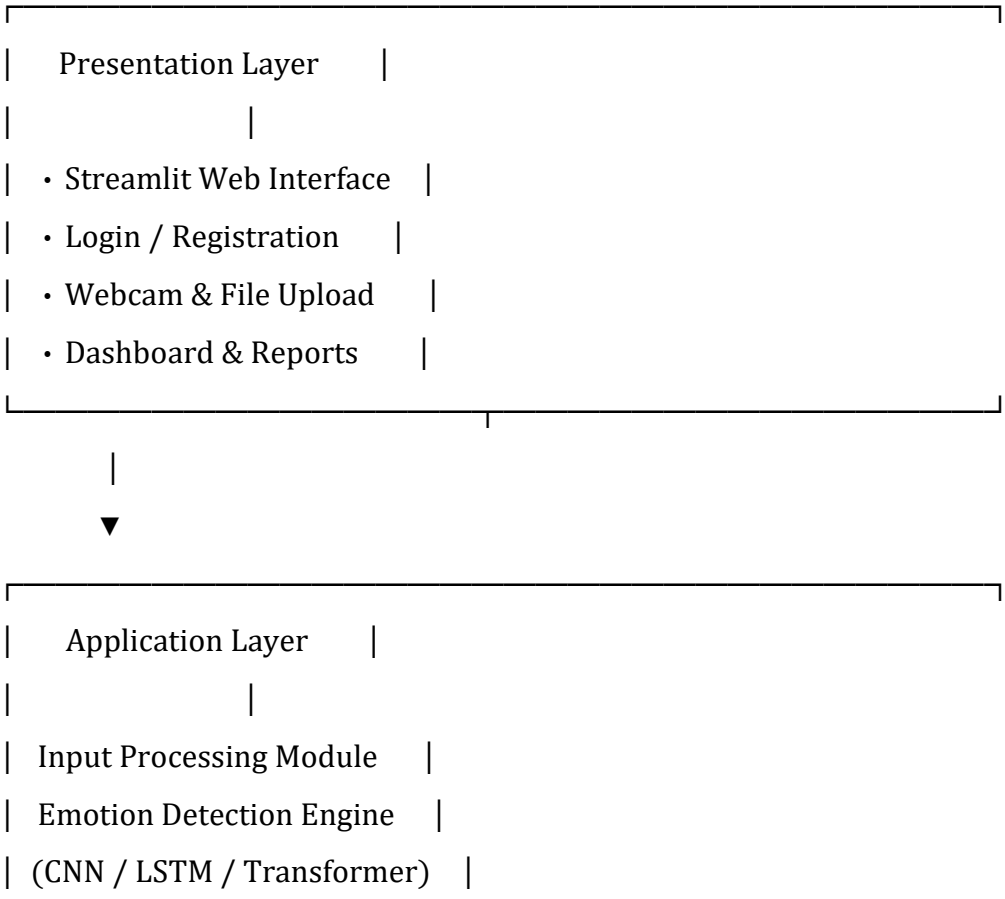
Solution Architecture

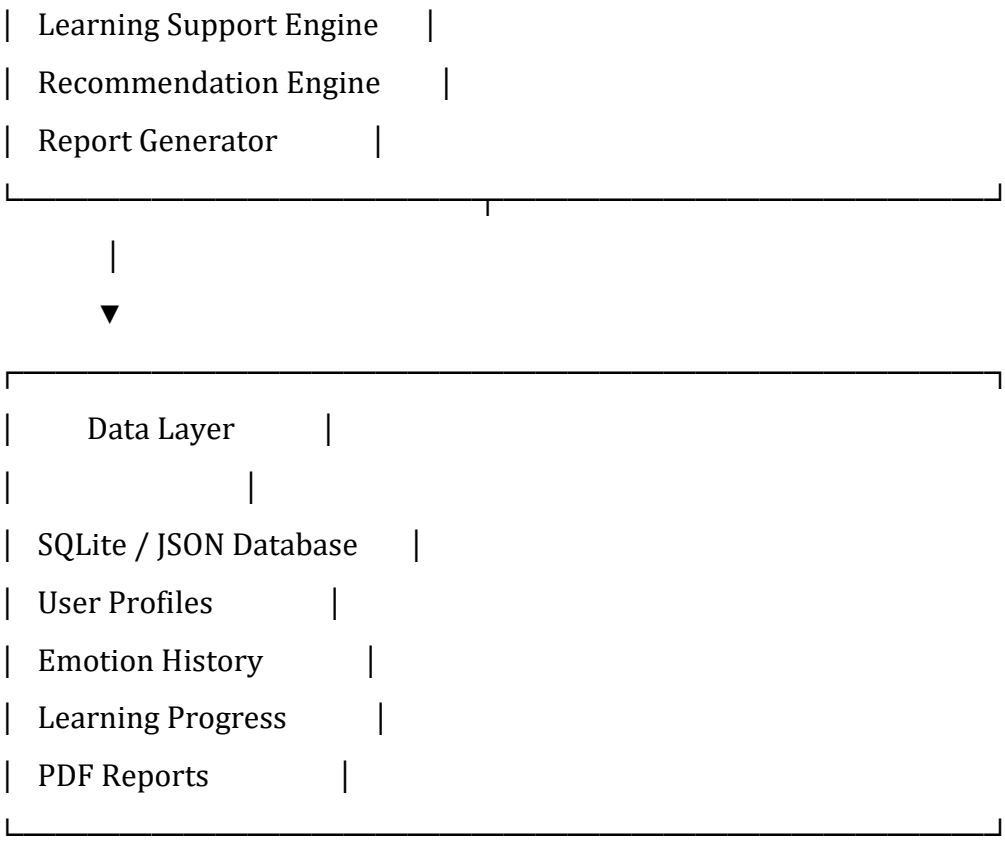
Date	2 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram

The **Emotion Detection and Learning Support Engine (EDLSE)** follows a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. User inputs such as facial images, voice recordings, or text are processed by AI models to detect emotions, generate personalized learning recommendations, and securely store reports and user progress.

Figure 2: Solution Architecture Diagram





Architecture Description

- **Presentation Layer:** Provides an interactive web interface built with **Streamlit**, allowing users to log in, upload facial images, voice recordings, or text, and view emotion analysis, recommendations, and reports.
- **Application Layer:** Processes user inputs using AI and deep learning models. The Emotion Detection Engine identifies the user's emotional state, while the Learning Support Engine generates personalized learning recommendations. A Report Generator creates downloadable PDF reports.
- **Data Layer:** Securely stores user information, detected emotions, learning progress, recommendation history, and generated reports using **SQLite/JSON** storage, enabling progress tracking and future analysis.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for audio upload, report viewing and interaction.	HTML5, CSS3, Streamlit
API Gateway	Routes requests between the client interface and AI services.	Python, Streamlit
Core Logic Service	Performs transcription, semantic similarity analysis, scoring and report generation.	OpenAI Whisper, Sentence Transformers, Librosa, FPDF
Database / Storage	Stores reference concepts, generated reports and uploaded audio.	JSON, Local Storage